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Iono-PhyTAU: A Lightweight Global Ionospheric TEC Prediction Model Based on Temporal Attention Unit and Physical Condition Constraints of the Ionosphere

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Key Points:

- The Iono-PhyTAU model effectively captures long-range spatiotemporal dependencies and mitigates error accumulation in multi-step prediction
- Compared with the sub-optimal model SA-ConvLSTM, Iono-PhyTAU reduces the RMSE by 14.5% in 2015 and 17.6% in 2020
- Iono-PhyTAU achieves 79.7% lower FLOPs compared with SA-ConvLSTM and improves TEC prediction under geomagnetic storm conditions

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Abstract Ionospheric Total Electron Content (TEC) is the primary error source causing signal delay in Global Navigation Satellite Systems (GNSS). High-precision TEC prediction is essential for positioning, navigation, and space weather monitoring. Although deep learning models including ConvLSTM, ConvGRU, and Transformer have advanced TEC forecasting, they still struggle to capture long-range global spatiotemporal dependencies, suffer from severe error accumulation in multi-step prediction, and involve high computational cost and large parameters. To address these limitations, this paper proposes a lightweight deep learning model called Iono-PhyTAU for global ionospheric TEC prediction. The model integrates a Temporal Attention Unit (TAU) and external physical driving constraints, which reduces computational complexity, enhances long-sequence dependence modeling, and extracts fine-grained spatiotemporal features. The model significantly improves TEC prediction accuracy during geomagnetic storms. Experiments conducted in the high solar activity year (2015) and low solar activity year (2020) show that Iono-PhyTAU achieves the best prediction accuracy, with RMSE values of 2.97 TECU and 1.08 TECU, respectively. The model effectively alleviates error accumulation in multi-step prediction: after the fourth hr of daily forecasting in 2015, its RMSE is reduced by 0.3–0.5 TECU compared with the suboptimal ConvLSTM model and by 0.5–1.5 TECU compared with C1PG. In addition, Iono-PhyTAU exhibits strong robustness across different latitudes, seasons, and times, with particularly outstanding performance during geomagnetic storms. Experimental results confirm that the model can adapt to various space weather environments and performs excellently in geomagnetic storm events. It has great application potential in improving the reliability of global TEC forecasting and provides strong support for GNSS ionospheric delay correction.

Plain Language Summary The ionosphere affects the accuracy of satellite navigation and communication systems. High-precision forecasting of ionospheric Total Electron Content (TEC) is critical for reducing signal errors and supporting space weather monitoring. However, existing deep learning models for global TEC prediction often suffer from large errors, fast error growth over time, and high computation costs. They also perform poorly during strong geomagnetic storms due to the lack of physical constraints. This study presents a lightweight and efficient model called Iono-PhyTAU for global ionospheric TEC prediction. The model uses a new Temporal Attention Unit (TAU) to capture long-term changes and reduce error accumulation, while combining real physical conditions (such as solar and geomagnetic activity) to improve prediction stability. Tests under high and low solar activity show that Iono-PhyTAU achieves higher accuracy than traditional models and significantly reduces prediction errors, especially during geomagnetic storms. The model is lightweight and computationally efficient, making it suitable for real-time global TEC forecasting and supporting reliable satellite navigation applications.

1. Introduction

Total Electron Content (TEC) is a critical parameter for characterizing ionospheric variations, which is widely used for satellite ionospheric delay correction, space weather monitoring, and communication systems (Liu et al., 2021, 2022; Rovira-Garcia et al., 2020). Accurate prediction of ionospheric TEC is essential for calculating ionospheric delays and holds significant value in services related to the Global Navigation Satellite System (GNSS), such as positioning, navigation, and timing (Feltens, 2007; Rovira-Garcia et al., 2015; Tang et al., 2022). However, due to the influence of various factors, including solar activity, geomagnetic activity, and atmospheric

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disturbances, the spatiotemporal variations of ionospheric TEC exhibit complex characteristics, posing significant challenges to the construction of accurate prediction models (Cander, 2016).

Since the establishment of the IGS Ionosphere Working Group, eight IGS analysis centers currently remain active in GIM-related work (Ogwala et al., 2022), among which the Center for Orbit Determination in Europe (CODE), Chinese Academy of Sciences (CAS), and Universitat Politècnica de Catalunya (UPC) have provided global ionospheric prediction products (García-Rigo et al., 2011; Li et al., 2018; Monte-Moreno et al., 2022). The products enable ionospheric Total Electron Content (TEC) forecasting with lead times of 1–48 hr, with the aim of establishing reliable long-term TEC prediction models to mitigate ionospheric adverse effects. Therefore, developing high-precision ionospheric TEC prediction models becomes crucial.

The methodologies employed for TEC prediction include physical models, empirical models, conventional statistical methods, and deep learning models. Physical models typically derive predictions from fundamental equations governing ionospheric plasma continuity, energy, and momentum (Fuller-Rowell et al., 1987; Schunk et al., 1986; Xiong et al., 2021). While physical models exhibit relatively high accuracy and provide substantial physical insights into the ionosphere/thermosphere system, of such models limitations have been acknowledged (Materassi, 2020). Physical models require precise boundary and initial condition specifications, demand substantial computational resources to solve partial differential equations and deliver qualitative results, and necessitate prolonged runtime to achieve stable and accurate outputs (Feltens et al., 2011).

Empirical models, such as the International Reference Ionosphere (IRI) model (Rawer et al., 1978) and NeQuick2 model (Nava et al., 2011), are widely adopted for TEC estimation and long-term forecasting. However, because empirical models primarily represent the average state of the ionosphere, they struggle to capture short-term variations in ionospheric dynamics, generally resulting in lower prediction accuracy (Chen et al., 2023; Xue et al., 2025). Statistical approaches, including the Adaptive Autoregressive (AAR) algorithm (Wang et al., 2018), linear time-series models (Suraj et al., 2018), and Autoregressive Integrated Moving Average (ARIMA) methods (Tang et al., 2020), have also been applied to TEC prediction by extracting temporal features from TEC time series. Nevertheless, the statistical methods face significant challenges in achieving high-precision forecasts due to statistical methods reliance on assumptions of stationarity and seasonality—conditions that may not consistently hold for irregularly varying ionospheric TEC data.

Recent advancements in artificial intelligence technology have motivated numerous researchers to investigate deep learning approaches for ionospheric TEC prediction (Lin et al., 2022). One of the most widely applied models is the Long Short-Term Memory (LSTM) network (Liu et al., 2020; Wen et al., 2021), which effectively alleviates the gradient vanishing problem in recurrent neural networks. Xiong et al. (2022) tested the LSTM method across 15 monitoring stations in China and found the performance of LSTM method superior to ARMA and IRI-2016 models. Another more advanced variant—the Bidirectional Long Short-Term Memory (Bi-LSTM) network, Bi-LSTM has demonstrated superior performance in TEC prediction by simultaneously leveraging both past and future temporal information. Vankadara et al. (2023) conducted a comparative study in the low-latitude region of Guntur, India, and the results showed that the Bi-LSTM model outperformed the conventional LSTM model across key metrics. Similar approaches include Gated Recurrent Unit (GRU) networks and variations of GRU networks.

In recent years, deep learning-based TEC forecasting has gradually evolved from station-based time-series forecasting to regional and global spatiotemporal map forecasting. LSTM, Bi-LSTM, and GRU models can characterize the nonlinear temporal variations of TEC, but they cannot explicitly model spatial interactions among different grid cells or geographical regions. To overcome this limitation, ConvLSTM, ConvGRU, and various attention-based models have been proposed to jointly extract spatial and temporal dependencies from sequential TEC maps (L. Liu et al., 2022; Q. Liu et al., 2022; Gao & Yao, 2023). These studies have demonstrated that incorporating spatial information can substantially improve TEC forecasting performance, particularly for global TEC map forecasting and under disturbed ionospheric conditions.

However, spatiotemporal forecasting models such as ConvLSTM and ConvGRU rely only on stacked convolutional operations and subsequently transform spatiotemporal features into forecast TEC maps through multiple fully connected layers. As the network depth increases, this approach inevitably leads to the loss of correlations between the current inputs and high-level contextual information. In addition, conventional stacked

convolutional architectures have difficulty effectively modeling long-range spatial dependencies during training, resulting in increasingly blurred forecast results.

In recent years, related research has no longer been limited to stacking standard convolutional and recurrent units but has shifted toward feature-extraction methods that focus on long-range dependencies, multiscale representations, and computational efficiency. Luo et al. (2023) applied SA-ConvLSTM, which incorporates a self-attention memory (SA) module, to global TEC forecasting. Compared with the original ConvLSTM, SA-ConvLSTM improves the extraction of global spatiotemporal features and produces more stable and accurate TEC forecasts. Xue et al. (2025) proposed RA-ConvLSTM, which combines a recurrent architecture with an attention mechanism to improve the modeling of long-range spatial dependencies. The CGAOA-STRA-BiConvLSTM framework proposed by Liu, Wang, et al. (2025) further improves the accuracy and robustness of global TEC prediction by combining the chaotic mapping and Gaussian mutation arithmetic optimization algorithm (CGAOA) with a spatio-temporal residual attention (STRA) module and a BiConvLSTM network. Li et al. (2025) proposed Ionformer, which employs a Transformer architecture with patch embedding and learnable positional encoding for global TEC forecasting. These studies indicate that recent TEC forecasting research has increasingly focused on long-range dependency modeling, fine-grained feature preservation, and more efficient global spatial representations.

Recent studies have also explored spatially aware feature extraction and wavelet- and frequency-domain feature extraction. Ma et al. (2025) proposed ED-CGConvLSTM, which incorporates CoordGate into ConvLSTM to achieve position-dependent spatial feature extraction and improve the representation of complex global TEC distributions. Zhou, Wu, et al. (2025) introduced denoising wavelet transform convolution to extract the low- and high-frequency features of TEC while suppressing noise. Zhou, Liu, et al. (2025) further developed CSA-WTConvLSTM by combining wavelet transform convolution with channel and spatial attention. In addition, Sun et al. (2026) proposed ST-TFFM-Observer to jointly model temporal- and frequency-domain dependencies within a dynamic-system framework. Xiao et al. (2026) proposed ED-FSA-DConvLSTM, which integrates deformable convolution, long-range spatial memory, and a lightweight design, thereby preserving long-range spatial information while substantially reducing memory consumption. Representative recent studies are summarized in Table 1.

The studies summarized in Table 1 show that current research focuses on spatially aware feature extraction, multiscale and frequency-domain decomposition, attention-based long-range dependency modeling, and lightweight network design. Nevertheless, global TEC forecasting still faces the challenge of simultaneously addressing the following limitations: effectively modeling long-range spatiotemporal dependencies, suppressing error accumulation during multi-step forecasting, improving forecasting performance during geomagnetic disturbances, and reducing computational complexity. Existing methods generally focus on addressing only some of these problems, whereas the joint treatment of these multiple objectives requires further investigation.

To address the limitations of the aforementioned spatiotemporal models, this paper introduces a Temporal Attention Unit (TAU) and incorporates physical driving constraint mechanisms to construct the prediction model. The TAU effectively alleviates the error accumulation issue in multi-step TEC prediction while featuring a lightweight structure and low computational cost (The computational cost is only 19.43 G FLOPs, representing an approximately 79.7% reduction compared with 95.55 G FLOPs of the sub-optimal SA-ConvLSTM model). Computational complexity is treated as a deployment constraint rather than a determinant of statistical prediction accuracy. The TAU module contains both intra-frame static attention and inter-frame dynamic attention. The static attention captures long-range dependencies within each TEC frame by acquiring an enlarged receptive field; the dynamic attention learns channel attention weights through the squeeze-and-excitation mechanism to model the temporal evolution of TEC along the time sequence. The final attention representation is obtained by fusing dynamic attention and static attention.

Meanwhile, a novel differential dispersion regularization method is embedded into the model to optimize the loss function for spatiotemporal prediction learning, which considers both the intra-frame prediction errors and the inter-frame variation magnitudes of TEC data. The proposed model fully utilizes the dynamic variation patterns and periodic correlation characteristics of ionospheric TEC, and achieves excellent TEC prediction performance under both geomagnetically quiet and disturbed conditions.

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Table 1
Representative Recent Deep Learning Models for Global Ionospheric TEC Prediction

Study	Model	Main feature-extraction strategy	Principal technical focus
Luo et al. (2023)	SA-ConvLSTM	Self-attention memory module integrated into ConvLSTM	Capturing global spatiotemporal dependencies and improving the stability and accuracy of global TEC forecasting
Xue et al. (2025)	RA-ConvLSTM	Recurrent architecture and attention mechanism	Modeling long-range spatial dependencies and reducing spatial-feature loss
Liu, Wang, et al. (2025)	CGAOA-STRA-BiConvLSTM	Multigranularity spatiotemporal feature retention and intelligent optimization	Preserving fine-grained features and optimizing model hyperparameters
Li et al. (2025)	Ionformer	Transformer, patch embedding, and learnable positional encoding	Modeling long-range dependencies in global TEC sequences
Ma et al. (2025)	ED-CGConvLSTM	CoordGate-based spatially aware convolution	Enhancing spatial-position perception and representing complex TEC distributions
Zhou, Wu, et al. (2025)	DWTConv-based model	Denoising wavelet transform convolution	Extracting low- and high-frequency features while suppressing noise
Zhou, Liu, et al. (2025)	CSA-WTConvLSTM	Wavelet convolution with channel and spatial attention	Jointly extracting low- and high-frequency features and emphasizing important spatial regions
Sun et al. (2026)	ST-TFFM-Observer	Time–frequency fusion and dynamic-system modeling	Jointly modeling temporal and frequency dependencies
Xiao et al. (2026)	ED-FSA-DConvLSTM	Frequency-domain self-attention and deformable convolution	Preserving long-distance spatial dependencies while reducing memory consumption

The primary objective of this study is to mitigate multi-step prediction error accumulation and incorporate external solar and geomagnetic drivers to improve the accuracy and robustness of one-day global TEC forecasting. Since operational global forecasting requires repeated generation of global TEC maps, computational efficiency serves as a secondary yet practically important objective. Accordingly, Iono-PhyTAU is designed to balance predictive performance and computational complexity.

To thoroughly evaluate the effectiveness of the prediction model proposed in this study, we compared Iono-PhyTAU with three deep learning baseline models (ConvLSTM, ConvGRU, and SA-ConvLSTM) trained and tested under the same protocol, as well as the operational CODE one-day forecast product C1PG. C1PG is not a deep learning model; it was included as a practical external benchmark for the same global TEC forecasting target. In Section 3, we performed comprehensive evaluation and analysis of the models, systematically comparing the prediction accuracy of the models under varying solar activity levels, latitudinal regions, different times of day, seasonal variations, and extreme space weather events.

2. Data and Methods

2.1. Ionospheric Data

This study uses global ionospheric TEC maps provided by the International GNSS Service (IGS), namely IGSG. The IGSG maps are distributed in the IONosphere Map EXchange (IONEX) format, a standard format developed by the IGS for exchanging and distributing ionospheric map data. This format facilitates the sharing of global TEC products and improves data interoperability among research institutions and analysis centers. Owing to its high accuracy and reliability (Hernández-Pajares et al., 2009; Roma-Dollase et al., 2018), IGSG is widely used as reference data for validating ionospheric TEC map prediction models (Li et al., 2024; Liu, Li, et al., 2025; Xia et al., 2022). The time resolution of this data is 2 hr, the longitude ranges from 180° west longitude to 180° east longitude with a resolution of 5°, the latitude ranges from 87.5° north latitude to 87.5° south latitude with a resolution of 2.5°, and the scale of the global TEC map grid points is 71 × 73. To assess the predictive

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Table 2
Training Set, Validation Set, and Test Set Partitioning

Solar activity	Training	Validation	Test
High	2011.01.01–2014.6.30	2014.07.01–2014.12.31	2015.01.01–2015.12.31
Low	2016.01.01–2019.6.30	2019.07.01–2019.12.31	2020.01.01–2020.12.31

performance of the model under different levels of solar activity, this study selects 5-year high solar activity data (1 January 2011 - 31 December 2015) and 5-year low solar activity data (1 January 2016 - 31 December 2020) within the study area. This study uses the global historical TEC data of the previous day (a total of 12 maps with a time resolution of 2 hr) as the input for a single sample to predict 12 global TEC data maps of the next day in advance. This prediction is a global TEC diurnal variation forecast based on IGSG data. The input dimension of the ionospheric TEC map is (batchsize, 12, 1, 71, 73), where batchsize represents the number of selected samples for a single training session, 12 is the length of the ionospheric TEC data of the previous day, 71 and 73 are the width and height of the ionospheric TEC grid, respectively, and the number of channels is 1. Additionally, this study employs a rolling window methodology, advancing by a one-day interval with each iteration. This study uses samples from two periods (2010–2014, 2016–2019) as training and validation samples, while the remaining two years (2015, 2020) are used as testing samples. The final distribution of samples is shown in Table 2.

2.2. Iono-PhyTAU Model

As shown in Figure 1, the global ionospheric TEC prediction model proposed in this paper adopts an encoder-decoder spatiotemporal fusion architecture, which can accurately capture the complex spatiotemporal evolution of ionospheric electron content. Meanwhile, it constrains and co-drives TEC features through external physical conditions, thereby improving the prediction robustness of the model under various space weather conditions. The model is constructed based on the encoder-decoder architecture, in which four standard two-dimensional convolutional layers are employed to constitute the spatial encoder, which perform multi-scale

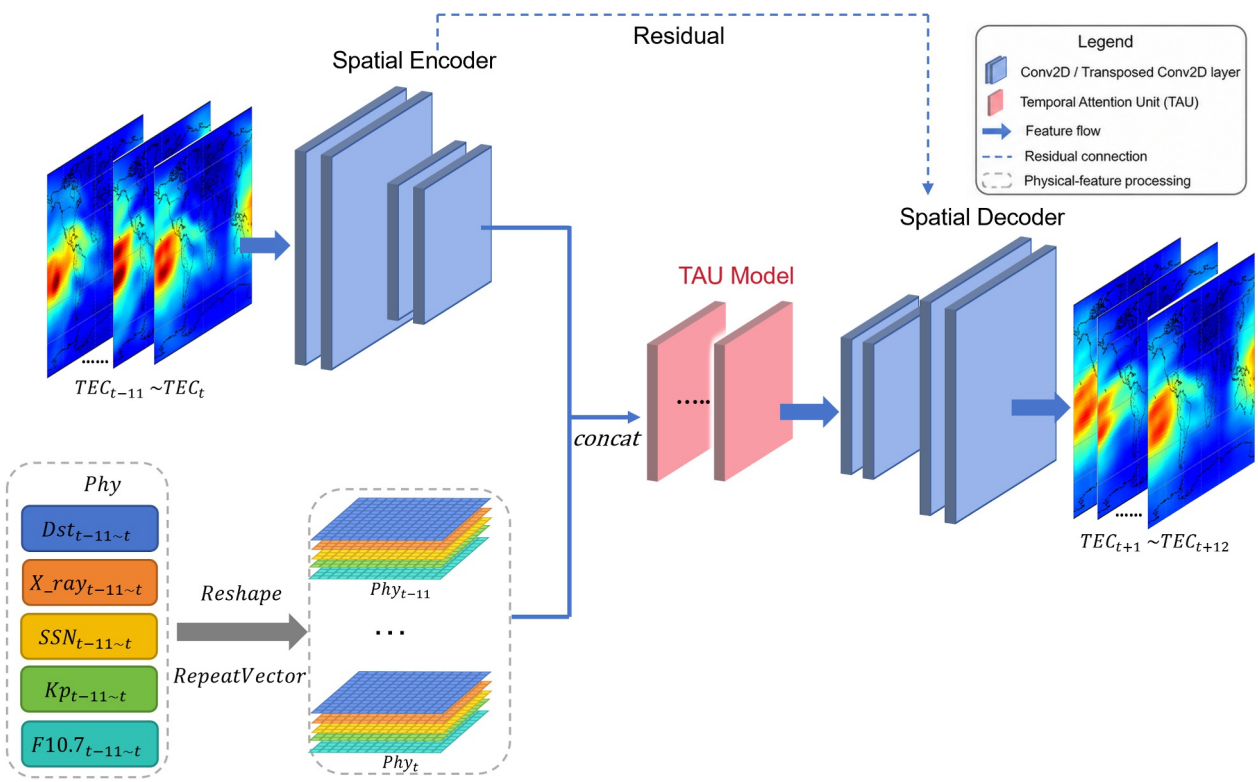


Figure 1. The overview architecture of our proposed model.

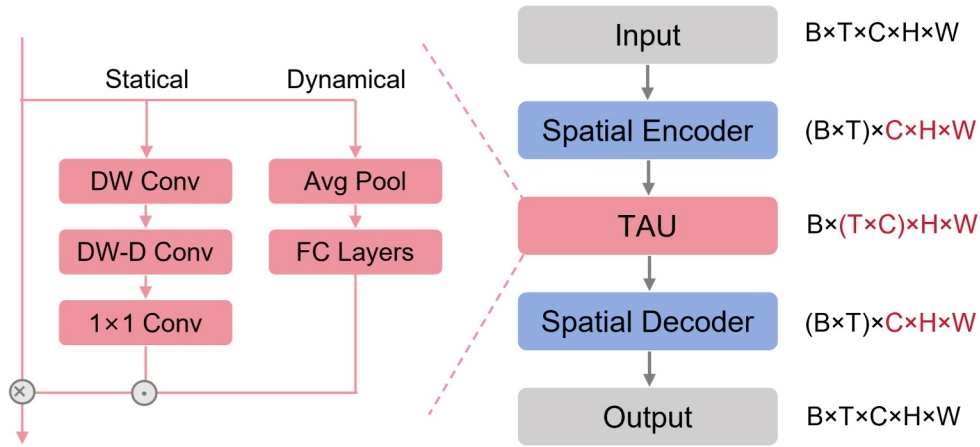


Figure 2. The detailed schema of our model.

spatial feature extraction on the input sequential TEC grid sequence ($TEC_{t-11} \sim TEC_t$) to effectively characterize the global spatial distribution patterns of the ionosphere (e.g., equatorial anomaly, polar disturbances, etc.); correspondingly, the spatial decoder employs four 2D transposed convolutional layers to gradually restore spatial resolution and generate 12-step ahead TEC prediction results ($TEC_t \sim TEC_{t+12}$).

To address the loss of fine-grained spatial features caused by deep convolution, a residual connection is established between the first convolutional layer of the spatial encoder and the last transposed convolutional layer of the decoder. This directly preserves the original spatial-dependent features while alleviating the gradient vanishing problem, ensuring the stability of model training. At the core of spatiotemporal feature fusion, multiple stacks of TAU modules are deployed between the encoder and decoder to capture the long- and short-term temporal evolution of ionospheric TEC, replacing traditional recurrent architectures for efficient temporal dependency modeling.

Considering the physical nature that ionospheric TEC variations are directly driven by solar and geomagnetic activities, this model innovatively adds a space weather physical parameter fusion branch at the output of the spatial encoder. Through dimension reshaping and repeated mapping operations, this branch performs dimension adaptation on key physical prior parameters (geomagnetic indices Dst, Kp, solar radiation index F10.7, X-ray flux, sunspot number SSN, etc.), and then concatenates and fuses them with the TEC spatial features extracted by the encoder to achieve constraint and driving effects. This fusion approach enables the model to deeply couple macro-scale external space weather drivers with local TEC spatial distribution characteristics, compensating for the lack of physical interpretability in purely data-driven models and effectively improving TEC prediction accuracy under extreme conditions such as geomagnetic storms and intense solar activity.

2.3. Temporal Attention Unit

As shown in Figure 2, the input to the model is a batch of sequential TEC grid maps, denoted by $I_{\mathrm{TEC}} \in R^{B\times T\times C\times H\times W}$, where B is the batch size, T is the number of time steps, C is the number of input channels, and H and W are the height and width of each TEC map, respectively. During spatial encoding and decoding, I_{TEC} is reshaped from $B \times T \times C \times H \times W$ *to* $(B \times T) \times C \times H \times W$. This operation temporarily merges the temporal and batch dimensions, allowing each TEC map to be processed independently by the shared spatial encoder and decoder. The network can therefore focus on extracting the global spatial distribution of TEC at each time step (i.e., the spatial structure of ionospheric disturbances). In the temporal feature modeling module, we reshape the feature dimension to $B \times (T \times C) \times H \times W$, and concatenate the TEC feature sequences at different moments along the channel dimension to realize the overall modeling of the TEC temporal evolution sequence.

As shown in Figure 3, we decompose the temporal attention mechanism for TEC into intra-frame static spatial attention and inter-frame dynamic temporal attention. We adopt small-kernel depth-wise convolutions (DW Conv), dilated depth-wise convolutions (DW-D Conv), and 1×1 convolutions to construct a large receptive field

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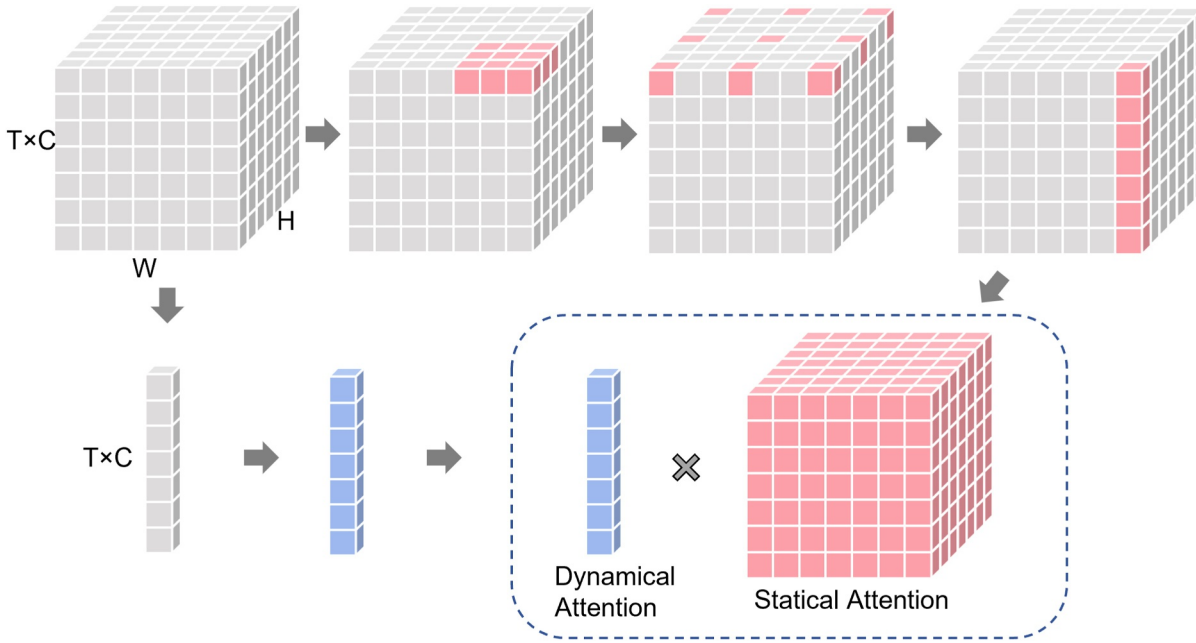


Figure 3. The intra-frame SA and the inter-frame DA architecture.

structure. With the global perception capability of the large receptive field, the intra-frame static attention can effectively capture long-range spatial dependencies among different latitude and longitude grid points and characterize the macroscopic spatial distribution of TEC. However, relying solely on static spatial features is insufficient to accurately fit the dynamic temporal evolution process of TEC. Therefore, we introduce an inter-frame dynamic attention mechanism, which adaptively learns the weight assignment of TEC feature channels at different moments in a squeeze-and-excitation manner. Finally, the dynamic temporal attention and static spatial attention are fused by multiplication to achieve accurate modeling of the spatiotemporal collaborative evolution characteristics of TEC.

Figure 2 illustrates the elaborate architectural design of our constructed model. The formulation of the devised TAU module is defined as Equation 1:

$$\begin{aligned} \mathrm{SA} &= \mathrm{Conv}_{1\times1}(\mathrm{DW}-\mathrm{DConv}(\mathrm{DWConv}(H))) \\ \mathrm{DA} &= \mathrm{FC}(\mathrm{AvgPool}(H)) \\ H' &= (SA \otimes DA) \odot H \end{aligned} \tag{1}$$

Here, $H \in R^{B\times(T\times C')\times H\times W}$ stands for the hidden feature tensor input into the TAU unit, while $\mathrm{SA} \in R^{B\times(T\times C')\times H\times W}$ and $\mathrm{DA} \in R^{B\times(T\times C')\times1\times1}$ correspond to static attention features and dynamic attention features respectively. FC refers to fully connected layers and AvgPool represents average pooling operation. In this work, $\otimes$ signifies the Kronecker product, and $\odot$ denotes the Hadamard product, both of which facilitate the refined feature fusion to generate high-precision predicted TEC maps.

To further improve the TEC prediction performance of the model, this paper introduces a differential divergence regularization method, which encourages the model to learn the variations between consecutive frames of global ionospheric TEC and capture the inherent variation patterns of ionospheric TEC. Given the predicted TEC maps $\hat{Y} \in R^{T'\times C\times H\times W}$ and the corresponding ground truth TEC maps Y, we first calculate their forward difference $\Delta\hat{Y}$ and $\Delta Y \in R^{(T'-1)\times C\times H\times W}$ as follows:

$$\begin{aligned} \Delta\hat{y}_i &= \hat{y}_{i+1} - \hat{y}_i \\ \Delta y_i &= y_{i+1} - y_i \end{aligned} \tag{2}$$

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Next, we convert the computed forward differences into probability distributions by applying the softmax activation across the channel, height, and width dimensions, yielding $\sigma(\Delta\ \hat{Y})$ and $\sigma(\Delta Y)$, which are defined as:

$$\sigma(\Delta\hat{y})_{i,j,k,l,t} = \frac{\exp(\Delta\hat{y}_{i,j,k,l,t}/\tau)}{\sum_{j'=1}^{C}\sum_{k'=1}^{H}\sum_{l'=1}^{W}\exp(\Delta\hat{y}_{i,j',k',l',\mu'}/\tau)}$$
$$\sigma(\Delta y)_{i,j,k,l,t} = \frac{\exp(\Delta y_{i,j,k,l,t}/\tau)}{\sum_{j'=1}^{C}\sum_{k'=1}^{H}\sum_{l'=1}^{W}\exp(\Delta y_{i,j',k',l',\mu'}/\tau)} \quad (3)$$

where τ denotes the temperature coefficient empirically set to 0.1 in order to sharpen the probability distribution. Via the competitive gating mechanism of the softmax function, penalties are deliberately imposed on TEC frames with large forward differences.

Thus, the differential divergence regularization term $\mathcal{L}_{\text{reg}}$ is formulated as the Kullback-Leibler (KL) divergence between the probability distributions of the predicted TEC maps $\sigma(\Delta\ \hat{Y})$ and the true TEC maps $\sigma(\Delta Y)$:

$$\mathcal{L}_{\text{reg}}(\hat{y},y) = D_{\text{KL}}(\sigma(\Delta\hat{y})\|\sigma(\Delta y))$$
$$= \sum_{i=1}^{T'-1}\sigma(\Delta\hat{y}_i)\log\frac{\sigma(\Delta\hat{y}_i)}{\sigma(\Delta y_i)}. \quad (4)$$

The proposed model is trained in an end-to-end unsupervised learning paradigm, and the total loss function integrates the mean square error loss and the differential divergence regularization term, with a weighting coefficient α assigned to the latter. This design contributes to generating more reliable predicted TEC maps.:

$$\mathcal{L} = \sum_{i=1}^{T'}\left\|\hat{y}_i - y_i\right\|^2 + \alpha\mathcal{L}_{\text{reg}}(\hat{y},y) \quad (5)$$

The former component concentrates on the discrepancies within individual frames, while the latter regularization constraint targets the variations occurring between consecutive frames, optimizing the precision of predicted TEC maps.

2.4. Implementation Details

All comparison models were rigorously optimized during the training phase through comprehensive hyperparameter tuning experiments to attain their optimal predictive performance for generating predicted TEC maps. The finalized hyperparameter settings are as follows: an identical 4-layer model cell structure with 64 hidden states, a batch size of 12, a learning rate of 0.0001, and 50 training epochs. The core objective of this unified configuration is to eliminate the interference of parameter discrepancies on model comparison. This paper acknowledges that independently adjusting parameters for distinct models can better accommodate their structural characteristics, which will be further verified in subsequent research. An early stopping mechanism with a patience value of five is adopted to automatically terminate training and save the optimal model when the validation error plateaus. The training procedure integrates scheduled sampling and layer normalization techniques, and is optimized via the Adam algorithm. In the encoding stage, each convolutional layer is followed by the LeakyReLU activation function with a negative slope coefficient of 0.01. The decoder maintains the same parameter configuration and applies the same activation function after each transposed convolutional layer, ensuring that the negative components of the data can compute small gradients through the LeakyReLU function, enabling continuous parameter updates and alleviating the vanishing gradient problem.

2.5. Statistical Indices

To comprehensively validate the performance of the Iono-PhyTAU model in predicting ionospheric TEC maps, we selected three classical spatiotemporal prediction models (ConvLSTM, ConvGRU, and SA-ConvLSTM) for comparison. The ConvLSTM model embeds convolutional layers within the LSTM architecture to enhance

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spatial data feature extraction. The ConvGRU model integrates gated recurrent units (GRU) with convolutional operations, maintaining spatiotemporal feature modeling capabilities while significantly reducing parameter quantities. The parameter count of the ConvGRU model (2.156 million) is approximately 25% less than that of ConvLSTM (2.875 million), achieving both computational efficiency and dynamic memory advantages. Whereas the SA-ConvLSTM employs a self-attention mechanism to dynamically capture global spatiotemporal dependencies, adaptively focusing on critical regional features, thereby effectively improving pattern recognition capability and prediction stability for complex ionospheric disturbances. While also incorporating the predicted GIM product C1PG (one-day forecast product from CODE). During the testing period, the prediction results of five models were rigorously evaluated against IGS-TEC (IGSG) data. Furthermore, we employed three Statistical Indices for model evaluation: Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Coefficient of Determination (R^2), and the mathematical formulations are as follows:

$$\begin{aligned} \mathrm{RMSE} &= \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2} \\ \mathrm{MAE} &= \frac{1}{n}\sum_{i=1}^{n}\left|\hat{y}_i - y_i\right| \\ R^2 &= 1 - \frac{\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2}{\sum_{i=1}^{n}\left(y_i - \bar{y}\right)^2} \end{aligned} \tag{6}$$

where y_i is the observations, $\hat{y}_i$ is the predictions, and $\bar{y}$ is the average of the observations, n is the total number of samples.

3. Results and Discussion

In the field of ionospheric TEC prediction using deep learning models, selecting a reliable reference benchmark is crucial for evaluating model performance. According to the evaluation by Roma-Dollase et al. (2018), when compared with altimeter VTEC data (from day 180 of 2001 to day 7 of 2016), IGSG has a standard deviation of 3.9 TECU and a relative error of 19.9%, and when compared with dSTEC data from independent GNSS stations, IGSG has a relative error of 28.9%. IGSG has been widely validated through extensive comparisons with multi-source observation data, and is widely recognized for its high precision and stability, serving as one of the primary ground truth data sets for model training, validation, and testing. Since the model is trained and tested with IGSG as the reference value, comparing with IGSG provides a reliable approach to directly assess the accuracy and learning capability of TEC prediction models. In this section, we conduct a comprehensive evaluation of the predictive performance of each model on the test set, including assessments under different solar activity, latitudes, time periods, months, and extreme space weather events.

3.1. Performance Under the Different Solar Activity Test Periods

This study selects the high solar activity year (2015) and the low solar activity year (2020) as research periods. The proposed Iono-PhyTAU model is comprehensively evaluated and compared with baseline models, including C1PG, ConvLSTM, ConvGRU, and SA-ConvLSTM. As shown in Table 3, Iono-PhyTAU achieves superior statistical metrics in both 2015 and 2020.

In 2020, under low solar activity conditions, the RMSE of Iono-PhyTAU reaches 1.08 TECU, the MAE is 0.74 TECU, and the coefficient of determination R^2 is 0.972, outperforming all comparison models. Compared with the suboptimal SA-ConvLSTM model, the RMSE and MAE of Iono-PhyTAU are reduced by 17.6% and 15.9%, respectively, which demonstrates that the proposed model possesses higher accuracy for global TEC map prediction. The high R^2 further indicates that the model can capture more than 97% of the variance in TEC dynamics and exhibits excellent consistency with the IGS reference data. Moreover, the model can effectively suppress extreme prediction outliers, which is crucial for high-precision space weather forecasting.

Comparative validation is further conducted in 2015, a high solar activity year characterized by intense ionospheric disturbances and dramatic TEC variations. As presented in Table 3, Iono-PhyTAU achieves state-of-the-

Table 3
Global TEC Prediction Results of all Models During Different Solar Activity Test Periods

	RMSE (TECU)	MAE (TECU)	R^2
High solar activity (2015)			
C1PG	4.23	2.98	0.932
ConvLSTM	4.21	2.98	0.933
ConvGRU	3.85	2.68	0.943
SA-ConvLSTM	3.47	2.36	0.954
Iono-PhyTAU	2.97	2.04	0.967
Low solar activity (2020)			
C1PG	1.61	1.15	0.940
ConvLSTM	1.51	1.07	0.947
ConvGRU	1.44	1.00	0.952
SA-ConvLSTM	1.31	0.88	0.960
Iono-PhyTAU	1.08	0.74	0.972

art performance with an RMSE of 2.97 TECU, an MAE of 2.04 TECU, and an R^2 of 0.963, surpassing all baseline models. In comparison with SA-ConvLSTM, the RMSE and MAE of Iono-PhyTAU decrease by 14.5% and 13.6%, respectively, reflecting its strong robustness in fitting complex TEC patterns under extreme space weather conditions. Even during geomagnetically active periods, the R^2 remains above 0.96, explaining over 96% of the TEC variations.

The RMSE distribution results during the high solar activity year in Figure 4 further verify the reliability of Iono-PhyTAU. For the proposed model, 63.3% of predictions have an RMSE lower than 3.0 TECU, while only 1.6% exceed 5 TECU. In contrast, traditional models such as C1PG and ConvLSTM show considerably higher proportions of large errors (23.0% and 18.4%, respectively), indicating their limited adaptability to severe ionospheric disturbances. Iono-PhyTAU exhibits stronger stability under low solar activity conditions. The RMSE distribution analysis in Figure 5 further confirms this stability: 50.3% of daily predictions by Iono-PhyTAU achieve RMSE values below 1.0 TECU (mainly within the 0.5–1.0 TECU range), with only 0.5% exceeding 2.0 TECU.

By comparing experimental results from high and low solar activity years, the proposed model is proven to be adaptable across the entire solar cycle. During low solar activity, the model accurately captures weak and subtle ionospheric disturbances. During high solar activity, the spatial feature extraction structure and temporal attention mechanism of Iono-PhyTAU effectively separate the nonlinear TEC variations dominated by solar flares and geomagnetic storms. Combined with physical constraint conditions, this dual adaptive characteristic enables Iono-PhyTAU to serve as a universal framework for global TEC forecasting. It can support high-reliability applications including satellite signal delay correction and real-time ionospheric storm monitoring, especially under extreme space weather scenarios.

To fully validate the effectiveness of physical constraint conditions, ablation experiments are carried out in this study. Table 4 presents the performance comparison after embedding different modules under high solar activity in 2015. The baseline Iono-TAU model achieves an RMSE of 3.40 TECU and an MAE of 2.37 TECU. After incorporating physical constraints, the prediction performance is significantly improved, with the RMSE decreasing to 2.97 TECU and the MAE dropping to 2.04 TECU, which greatly enhances the overall prediction accuracy.

To further verify the advances of Iono-PhyTAU, two recently-proposed global TEC prediction models, LARRES (Zhuang et al., 2025) and RA-ConvLSTM (Xue et al., 2025), are selected as advanced baselines. LARRES enhances the capability of capturing global ionospheric spatial features via a large-receptive-field convolutional structure. RA-ConvLSTM introduces a recurrent attention module on the basis of ConvLSTM to strengthen the modeling of long-range spatiotemporal dependencies. In this work, these two advanced models adopt exactly the same data set partition, preprocessing pipeline and hyperparameter settings as Iono-PhyTAU for fair and reasonable prediction-performance comparisons.

As shown in Table 5, Iono-PhyTAU achieves the best overall performance among the three models under both high and low solar activity conditions. In 2015, Iono-PhyTAU obtains an RMSE of 2.97 TECU, an MAE of 2.04 TECU, and an R^2 of 0.967. Compared with LARRES, RMSE and MAE are reduced by 8.3% and 3.8%, respectively, with an absolute R^2 improvement of 0.004. Relative to RA-ConvLSTM, Iono-PhyTAU reduces RMSE by 16.3% and MAE by 17.1%, while increasing R^2 from 0.955 to 0.967. These results indicate that Iono-PhyTAU can more effectively capture the complex and rapidly-varying global TEC structures under high solar activity.

The performance advantage becomes more prominent in 2020. Iono-PhyTAU yields an RMSE of 1.08 TECU and an MAE of 0.74 TECU, representing reductions of 18.2% and 17.8% compared with LARRES. Against RA-ConvLSTM, both RMSE and MAE decrease by 24.5%. Iono-PhyTAU also reaches the highest R^2 of 0.972, in contrast to 0.964 for LARRES and 0.959 for RA-ConvLSTM. It can be seen that the performance improvement is

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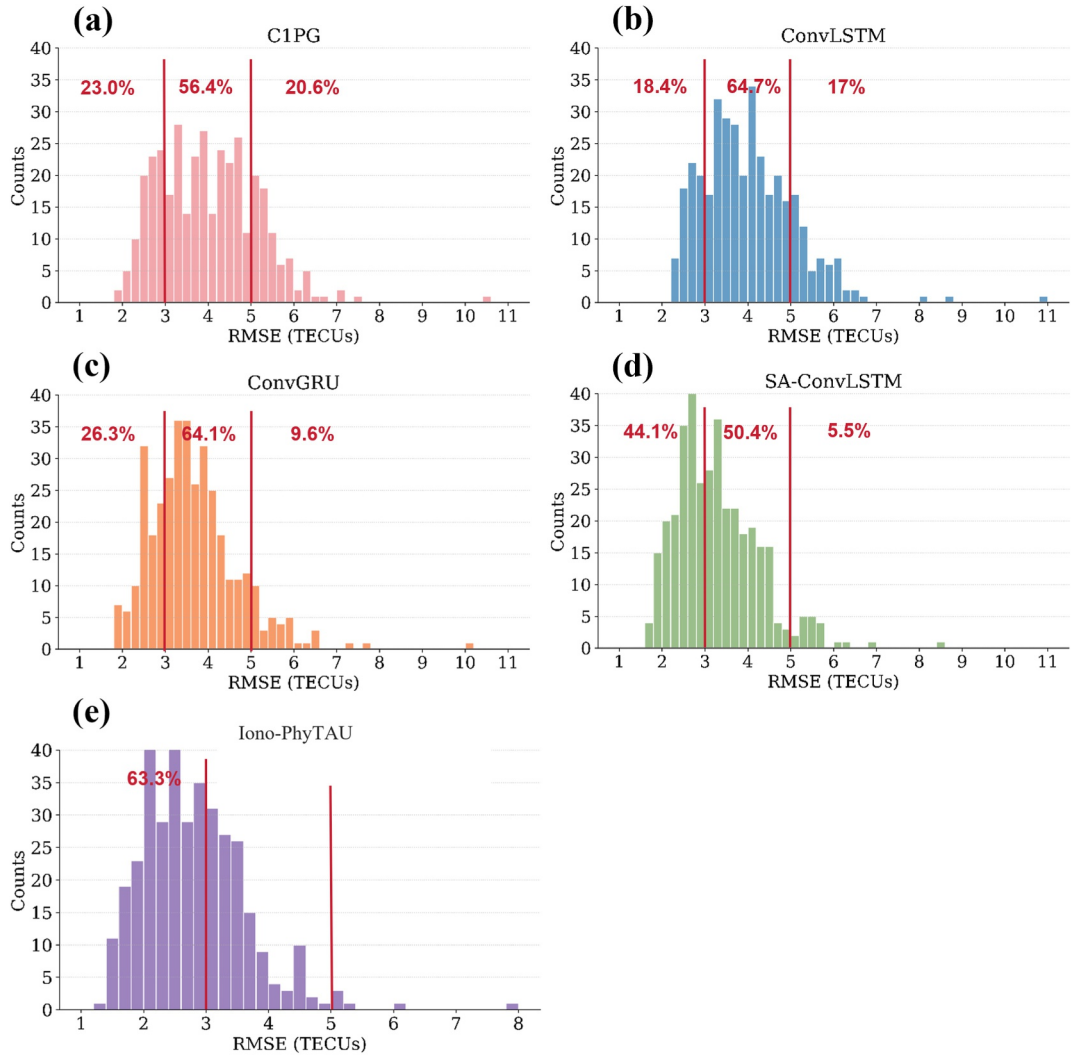


Figure 4. Distribution of daily mean RMSE values for the four prediction models and the C1PG during the high solar activity year (2015).

not limited to a single solar-activity level, demonstrating that Iono-PhyTAU maintains stable prediction capability under distinctly different ionospheric environments.

3.2. Performance Under the Different Latitudes

Figure 6 compares the RMSE and MAE of TEC predictions obtained from different models across various latitudinal regions. During the high solar activity year (2015), the Iono-PhyTAU model achieves the optimal error performance at all latitudes. Compared with the second-best model SA-ConvLSTM, the RMSE of Iono-PhyTAU is reduced by 0.2–0.9 TECU, and the accuracy improvement is more prominent in low-latitude regions. The spatial advantage becomes particularly evident during low solar activity year (2020). In contrast, the traditional C1PG model suffers obvious error accumulation in the southern high latitudes due to its insufficient ability to learn nonlinear features.

Table 6 statistically summarizes the RMSE, MAE, and coefficient of determination R^2 for different latitudinal zones. In low-latitude regions during the high solar activity year (2015), compared with SA-ConvLSTM, the proposed model reduces RMSE by 10.9% and MAE by 11.5%; compared with C1PG, the RMSE and MAE are decreased by 20.2% and 21.4%, respectively. In high-latitude regions under high solar activity, the RMSE and MAE reductions reach 35.8% and 35.4% relative to the C1PG model. In high-latitude regions during the low solar

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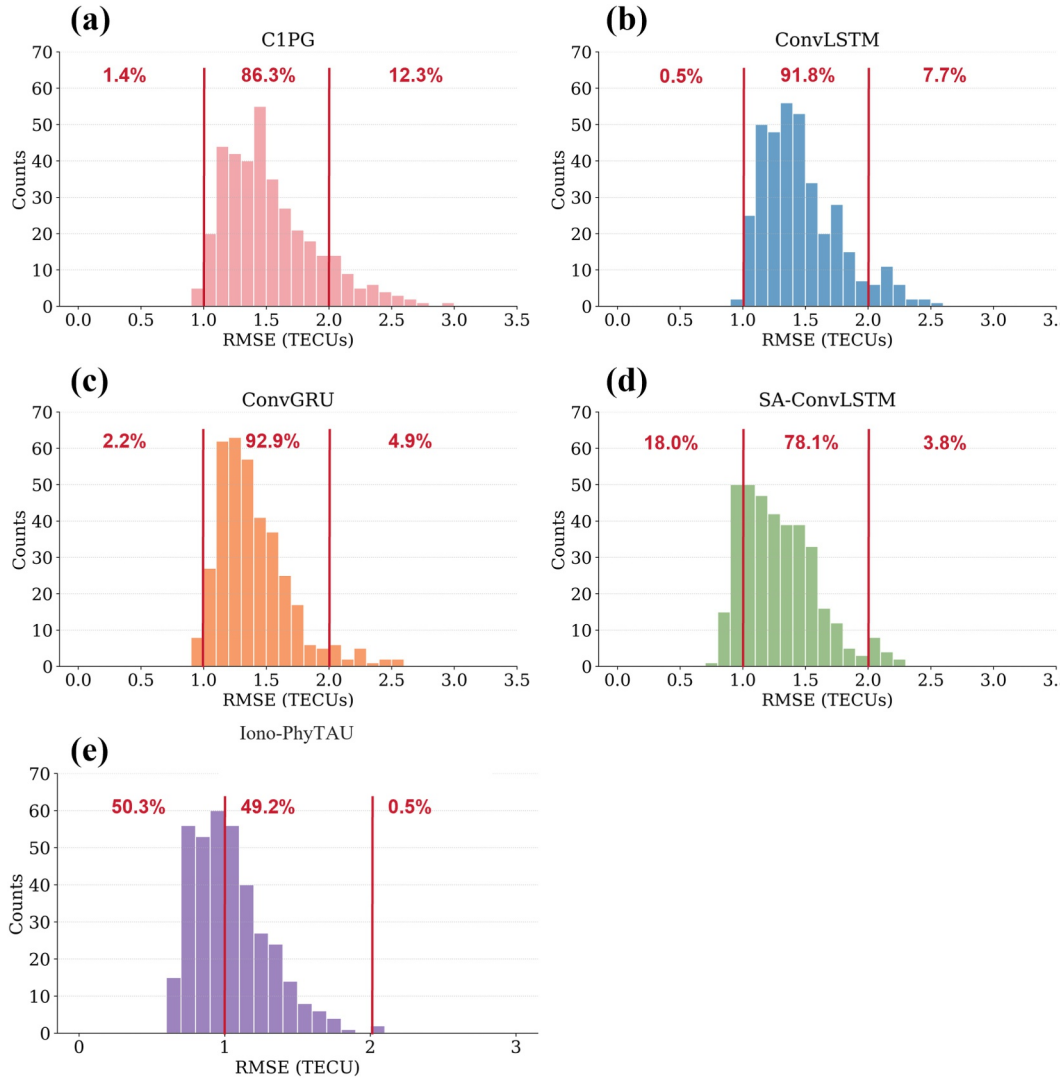


Figure 5. Distribution of daily mean RMSE values for the four prediction models and the C1PG during the low solar activity year (2020).

activity year of 2020, the Iono-PhyTAU model achieves an RMSE of 0.71 TECU and an MAE of 0.52 TECU, which are 46.2% and 46.4% lower than those of the C1PG model, respectively. Notably, the R^2 values of Iono-PhyTAU remain steadily above 0.89 across all latitudinal zones throughout the entire test period, demonstrating excellent and stable fitting performance under varying solar activity conditions.

3.3. Performance Under the Different Months and Different Times

Figure 7 shows the RMSE comparison results of TEC predictions by the five models across different months in 2015 and 2020. During the high solar activity year (2015), compared with the baseline models (C1PG, ConvLSTM, ConvGRU, and SA-ConvLSTM), Iono-PhyTAU achieves the lowest RMSE in all months. Affected by geomagnetic storm activity, the maximum prediction error of Iono-PhyTAU occurs in March. In March, its RMSE is reduced by 1.41 TECU (27.6%) compared with C1PG and by 0.69 TECU (15.9%) compared with SA-ConvLSTM. In terms of the annual average performance in 2015, the RMSE of Iono-PhyTAU decreases by 29.8% relative to C1PG and by 14.5% relative to SA-ConvLSTM. During the low solar activity year (2020), the Iono-PhyTAU model achieves the smallest overall error in June, with RMSE reduced by 0.51 TECU

Table 4
Comparison of the Influence of Integrating Physical Constraints on the Accuracy of TEC Prediction

Model (Iono-PhyTAU)	RMSE (TECU)	MAE (TECU)	R^2
Iono-TAU	3.40	2.37	0.955
Iono-PhyTAU	2.97	2.04	0.967

Table 5
Performance Comparison With Advanced Models Under the Unified Experimental Protocol

Test year	Model	RMSE (TECU)	MAE (TECU)	R^2
2015	LARRES	3.24	2.12	0.963
	RA-ConvLSTM	3.55	2.46	0.955
	Iono-PhyTAU	2.97	2.04	0.967
2020	LARRES	1.32	0.90	0.964
	RA-ConvLSTM	1.43	0.98	0.959
	Iono-PhyTAU	1.08	0.74	0.972

(40.8%) and 0.19 TECU (20.1%) compared with C1PG and SA-ConvLSTM, respectively.

Figure 8 presents the hourly RMSE comparison of TEC predictions from the five models in 2015. By evaluating model performance across different prediction horizons, this section investigates the temporal robustness of each method. A clear trend can be observed from Figure 8: the prediction accuracy of deep learning models gradually decreases as the forecast horizon increases. Among all models, the C1PG exhibits particularly large errors. In the first two hours of 2015, its RMSE remains between 4.0 and 5.0 TECU, making it the worst-performing model for global ionospheric model (GIM) prediction. In contrast, the Iono-PhyTAU maintains the lowest error level throughout the entire period, with its RMSE consistently below 3.0 TECU in 2015.

As the Earth rotates with the variation of Universal Time (UT), sunlight illuminates different geographical regions sequentially. This causes periodic fluctuations in global total electron content and the complexity of its spatio-temporal gradients with UT. For instance, at UT = 6:00, the Asia–Pacific region lies within local noon to early afternoon hours. In this period, the Equatorial Ionization Anomaly (EIA) develops intensely and exhibits complex structures. Meanwhile, the global total TEC and its spatial gradients reach their peaks, and the RMSE of global ionospheric prediction models also attains its maximum value accordingly (Lean et al., 2016; Ogwala et al., 2022; Olwendo et al., 2016; Zhang et al., 2022).

The magnitude and complexity of TEC in different longitudinal regions are controlled by multiple factors, including geomagnetic declination, the intensity of the Equatorial Electrojet (EEJ), and the meridional propagation of tides and planetary waves. The combined effects of these factors continuously change the TEC amplitude, spatial gradients, and EIA intensity within sunlit longitudinal sectors at different UT moments, thereby leading to variations in the prediction errors (e.g., RMSE) of global TEC models. Consequently, as shown in Figure 8, the prediction errors vary systematically with Universal Time and exhibit alternating local peaks and troughs.

In Figure 8, SA-ConvLSTM shows stable and competitive error performance at all time points and consistently ranks second. Therefore, this study focuses on comparing the prediction differences between SA-ConvLSTM and the proposed Iono-PhyTAU model. The two models achieve similar error levels at 00:00 UT; however, a significant divergence in prediction accuracy emerges starting from 04:00 UT each day. After 04:00 UT in 2015, the average RMSE difference between the two models reaches approximately 0.6 TECU. During the high solar activity year of 2015, the RMSE of SA-ConvLSTM at 04:00 increases by 1.05 TECU (42%) compared with that at 00:00, while the RMSE of Iono-PhyTAU increases by only 0.69 TECU (29.8%). At 22:00 UT, the RMSE of SA-ConvLSTM rises by 1.32 TECU (53%) relative to 00:00, whereas that of Iono-PhyTAU increases by merely 0.81 TECU (34%). During the low solar activity year of 2020, the model achieves a more significant improvement

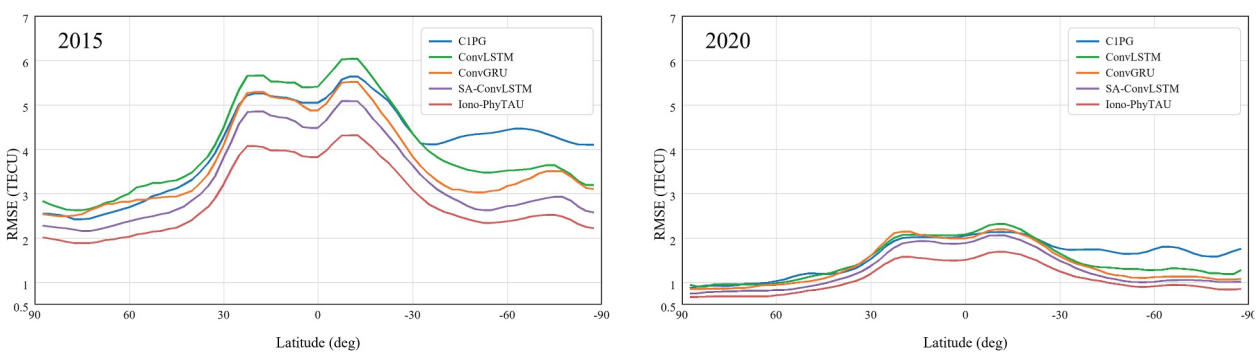


Figure 6. Average RMSE of the five prediction models compared with IGSG in different Latitudes of 2015 and 2020.

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Table 6
Global TEC Prediction Results of all Models During Different Latitudes Test Periods

	Latitude (°)	Low (−30, 30)			Middle (−60, −30]∪[30, 60)			High (−90, −60]∪[60, 90)		
	Model	RMSE (TECU)	MAE (TECU)	R^2	RMSE (TECU)	MAE (TECU)	R^2	RMSE (TECU)	MAE (TECU)	R^2
High Solar activity (2015)	C1PG	5.13	3.64	0.935	3.77	2.71	0.865	3.52	2.53	0.743
	ConvLSTM	5.44	3.94	0.927	3.54	2.58	0.880	3.13	2.32	0.797
	ConvGRU	4.98	3.52	0.939	3.13	2.24	0.907	3.00	2.21	0.813
	SA-ConvLSTM	4.59	3.23	0.948	2.81	1.94	0.925	2.53	1.84	0.867
	Iono-PhyTAU	4.09	2.86	0.959	2.52	1.74	0.940	2.26	1.64	0.894
Low Solar activity (2020)	C1PG	1.97	1.39	0.942	1.43	1.06	0.869	1.32	0.97	0.834
	ConvLSTM	2.03	1.45	0.938	1.22	0.92	0.905	1.04	0.80	0.897
	ConvGRU	1.99	1.42	0.941	1.13	0.84	0.918	0.93	0.69	0.919
	SA-ConvLSTM	1.82	1.24	0.950	1.00	0.72	0.935	0.84	0.64	0.933
	Iono-PhyTAU	1.47	1.03	0.967	0.87	0.63	0.952	0.71	0.52	0.952

in mitigating TEC prediction error accumulation. The RMSE of SA-ConvLSTM at 22:00 increases by 0.51 TECU (45%) compared with that at 00:00, while that of TCF-MSA-ConvLSTM increases by only 0.22 TECU (20%).

The results demonstrate that Iono-PhyTAU can maintain stable prediction accuracy even when the forecasting horizon is extended. This advantage benefits from the structural design of dynamic temporal attention and static spatial attention embedded in the TAU module. The analysis fully verifies that the proposed model possesses reliable and stable prediction capability for long-term forecasting scenarios.

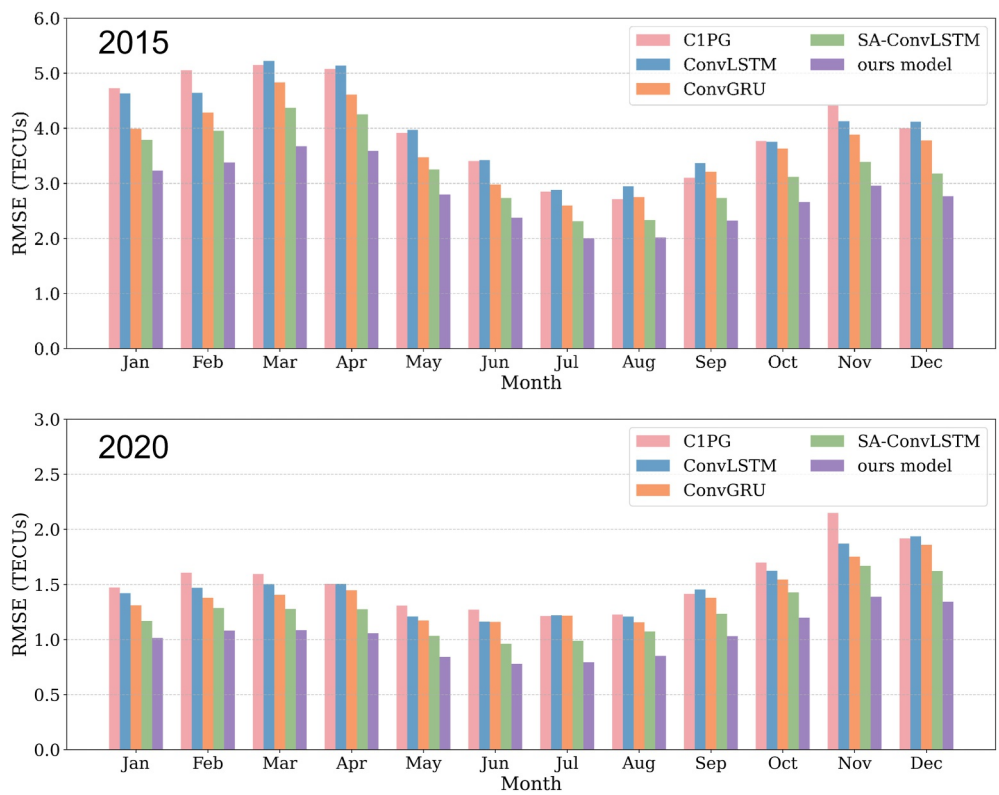


Figure 7. Average RMSE of the five prediction models compared with IGSG in different months of 2015 and 2020.

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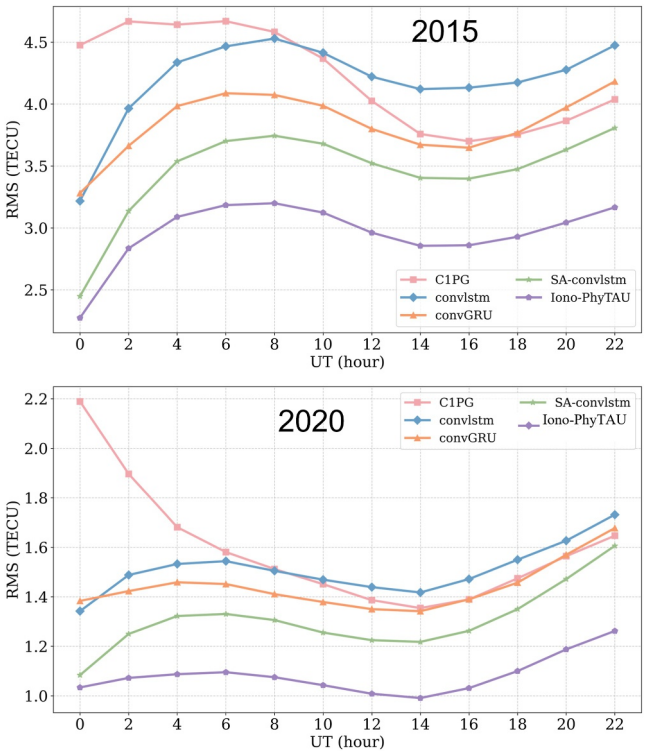


Figure 8. Average RMSE of the five prediction models compared with IGSG at 2-hr intervals (Universal Time) in 2015 and 2020.

3.4. Performance Under the Extreme Space Weather Events

Deep learning-based ionospheric TEC prediction models generally achieve satisfactory performance under normal space weather conditions. However, as reported by Chen et al. (2022), the prediction accuracy often decreases significantly during extreme space weather events such as solar flares, geomagnetic storms, and ionospheric storms. The main reasons for this performance degradation are the complex and rapidly evolving physical variations of the ionosphere during extreme events, as well as the incomplete understanding of the underlying physical mechanisms of space weather effects. Consequently, critical influencing factors cannot be fully incorporated into the modeling framework, inevitably limiting the prediction accuracy under extreme space weather conditions.

To evaluate the adaptability of the Iono-PhyTAU model under extreme space weather scenarios, a strong geomagnetic storm event on 10 April 2015 (Kp = 6) was selected for verification in this section. To intuitively demonstrate the capability of the Iono-PhyTAU model in suppressing the accumulation of prediction errors for ionospheric TEC maps, Figure 9 illustrates the global spatial error distribution between the TEC predictions of different models and the corresponding IGSG reference data. As the prediction lead time increases, the TEC prediction errors of all models accumulate globally to varying degrees. Overall, Iono-PhyTAU can effectively suppress the spatial accumulation of errors. Especially at 06:00 UT, the peak moment of the geomagnetic storm, the model shows a more prominent suppression effect near the equatorial region.

Furthermore, to further verify the stability and reliability throughout the entire geomagnetic storm period, Figure 10 compares the prediction deviations of all models relative to the IGS reference during the storm recovery phase. The results indicate that during the recovery stage, the MAE of Iono-PhyTAU is only 4.91 TECU, maintaining the smallest prediction deviation among all models and exhibiting highly consistent accuracy with that achieved during the main storm phase.

Geomagnetic storms trigger rapid and non-stationary variations in TEC. The evolution of ionospheric anomalies, such as sudden enhancements and depletions, is strongly dependent on the background ionospheric conditions before the storm. As geomagnetic activity intensifies, traditional models such as ConvLSTM suffer from weakened correlations between historical and current TEC data, resulting in delayed tracking and inaccurate fitting of anomalous evolutionary trends.

The proposed Iono-PhyTAU dynamically enhances temporal dependence through two core mechanisms. By combining intra-frame static attention and inter-frame dynamic attention, the model captures both long-range spatial dependencies within each TEC map and the temporal evolutionary trends across consecutive frames. This structure strengthens the representation of critical anomalous features and improves the continuity of TEC states during the pre-storm, storm peak, and post-storm stages.

In addition, TEC anomalies during geomagnetic storms exhibit significant spatial heterogeneity. Low-latitude equatorial anomaly regions contain both large-scale disturbances and small-scale irregular structures, making it difficult for baseline models such as ConvLSTM to simultaneously capture global anomalies and local fine-grained features. In contrast, the spatial encoder together with embedded physical constraints extracts multi-scale fine-grained features at different levels and fully integrates key physical influencing factors into the prediction modeling process, effectively alleviating the prediction challenges encountered during severe geomagnetic storm events.

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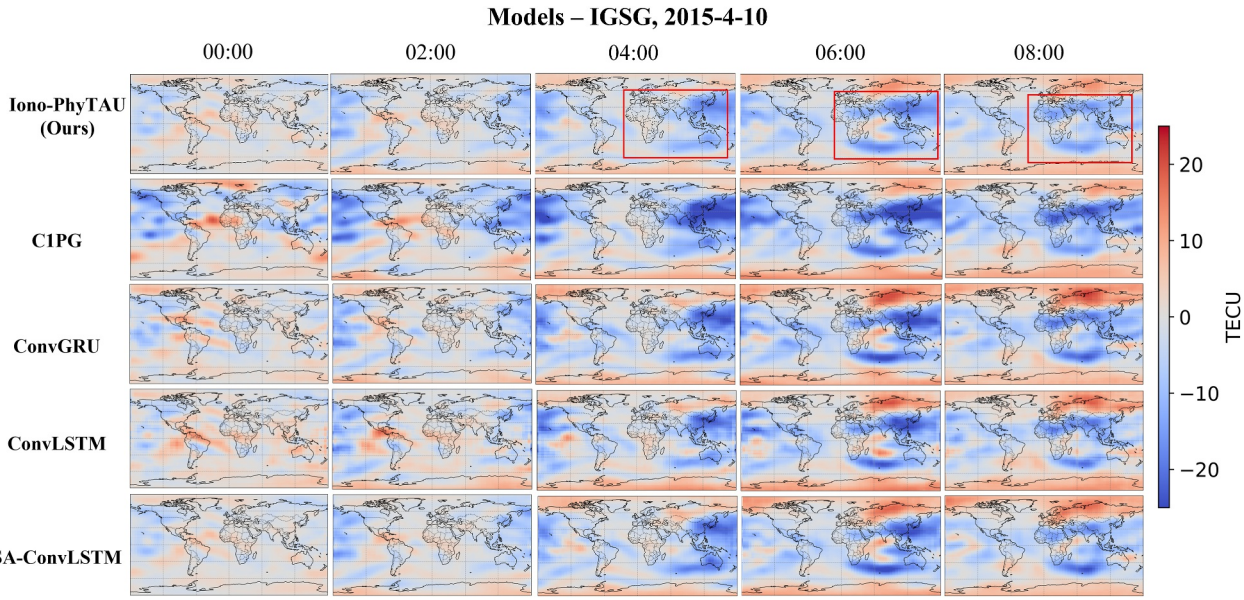


Figure 9. Global distribution of differences between the TEC predicted values of the five prediction models and the IGSG TEC reference values during the geomagnetic storm on 10 April 2015.

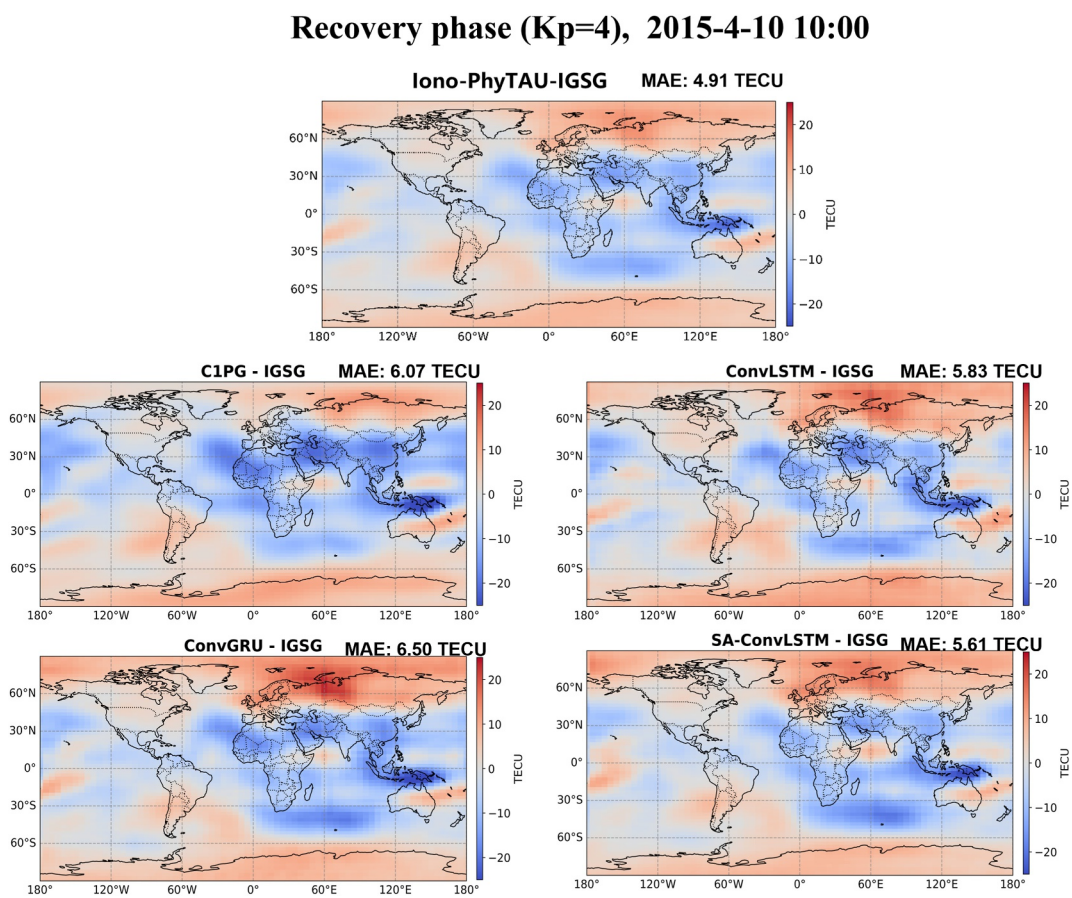


Figure 10. Global distribution of differences between the TEC predicted values of the five prediction models and the IGSG TEC reference values during the recovery phase of geomagnetic storms on 10 April 2015.

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4. Conclusions

The core reason for error accumulation in existing prediction models lies in their insufficient ability to capture long-term temporal dependencies and extract spatial features adequately. Based on the periodic variations and spatial correlation characteristics of historical TEC data, this paper integrates the TAU unit with spatial encoding to enhance the model's long-range temporal modeling capability and TEC spatial feature extraction performance, effectively compensating for the inherent limitations of current TEC prediction models. On this basis, physical constraint fusion is further employed for collaborative driving, which enhances the model's sensitivity during geomagnetic storms and improves prediction accuracy under storm conditions.

Extensive experimental results demonstrate that the proposed Iono-PhyTAU consistently outperforms classical models such as ConvLSTM, ConvGRU, and SA-ConvLSTM across different solar activity cycles, latitudes, and seasons. In particular, the model greatly alleviates error accumulation in multi-step TEC forecasting and achieves excellent prediction performance during geomagnetic storms. In the low solar activity year (2020), Iono-PhyTAU achieves an RMSE of 1.08 TECU and an MAE of 0.74 TECU, which are reduced by 17.6% and 15.9% compared with SA-ConvLSTM, respectively. In the high solar activity year (2015), its RMSE reaches 2.97 TECU, MAE is 2.04 TECU, and the coefficient of determination R^2 is 0.967, showing comprehensive superiority over all baseline models. Compared with SA-ConvLSTM, the RMSE and MAE of Iono-PhyTAU are decreased by 14.5% and 13.6%, respectively, while R^2 remains steadily at 0.967. Even during geomagnetic storms, the model can explain more than 96% of the variations in TEC. Spatially, Iono-PhyTAU outperforms baseline models across all latitude zones. In high-latitude regions in 2015, its RMSE and MAE are reduced by 35.8% and 35.4% relative to C1PG. Furthermore, monthly accuracy analysis shows that during the geomagnetically active period in March 2015, the RMSE of Iono-PhyTAU is 27.6% lower than that of C1PG and 15.9% lower than that of SA-ConvLSTM.

Temporally, Iono-PhyTAU exhibits the weakest error accumulation as the prediction lead time increases. Under extreme space weather events, the model achieves significant accuracy improvements in both low-latitude and high-latitude regions. The FLOPs of the proposed TEC prediction model are 19.43 G, accounting for only one-fifth of the computational cost of the suboptimal SA-ConvLSTM model (95.55 G). This study focuses on improving TEC prediction performance while maintaining a lightweight model structure. The Iono-PhyTAU achieves substantial accuracy gains under various solar activity conditions with low computational complexity, delivers outstanding performance during geomagnetic storms, and effectively mitigates error accumulation as the prediction step length increases.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

The IGSG data are openly accessible (https://cddis.nasa.gov/archive/gnss/products/ionex) from Crustal Dynamics Data Information System. The Kp index is available (https://datapub.gfz-potsdam.de/download/10.5880.Kp.0001) from GeoForschungsZentrum and Dst index is accessible (http://wdc.kugi.kyoto-u.ac.jp/dstdir/) from World Data Center for Geomagnetism, Kyoto. The related code and data of Iono-PhyTAU can be got from Jia & Gui. (2026).

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